

THE ℓ -COLUMN $\star$ -PIERI RULE ($\star\ell$) FOR $e_k(Y) \bullet (e_{a_1} \cdots e_{a_\ell})$, PROVED FOR ALL k, ℓ

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ABSTRACT. For every $m \geq 0$, $k \geq 0$ and $\ell \geq 1$ we prove a closed generating-function rule for $\sum_a \prod_c z_c^{a_c} t^{-\binom{k}{2}} e_k(Y) \bullet (e_{a_1} \cdots e_{a_\ell})$, as an identity in $\Lambda_m \otimes \mathbb{Q}(s, t)(z_1, \dots, z_\ell)$, $s = q^{-1}$. It specialises to Day 207b's $e_k \star e_r$ at $\ell = 1$ and to the two-column rule (TC) of Day 209 at $\ell = 2$. The proof is the Day 209 skeleton with ℓ poles instead of two; the only new input is a closing identity (Z), which is the residue theorem for one rational function.

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Status: Proved; self-checked and machine-checked; not yet reviewed by anyone else.
Prior reference: Two-column case (TC), notes/2026-09-29-two-column-TC-proved.pdf (“209”).

Please check: §5, Step B, the prefactor bookkeeping that produces $t^{2i_c - S} = A_c / \prod_{c' \neq c} A_{c'}$; and the pairwise splitting (**P6**) of κ'_c against the kernel ratio K_{I-e_c}/K_I . These are the two places where the $\ell = 2$ computation was generalised rather than copied.

1. STATEMENT

Conventions are those of 207b and Day 209 (proofs/2026-09-29-day209-two-column-TC-PROVED.md, “209” below):

- $s = q^{-1}$; $a_{ij} = (X_i - tX_j)/(X_i - X_j)$; $E(z) = \sum e_r z^r$;
- $c(n, j)$, $N_j^{(n)} = t^{-nj} c(n, j)$, $C_j(x) = \sum_n c(n, j) x^n$, D_j and G as in 207b §4;
- $K_{ij}(z, w)$ as in 209 §0, i.e. $K_{ij}(z, w) = \prod_{p < i} \frac{t^p z - sw}{t^p z - t^j w} \prod_{r < j} \frac{sz - t^r w}{t^i z - t^r w}$. It satisfies $K_{ij}(z, w) = K_{ji}(w, z)$: substitute $p \leftrightarrow r$ and flip both signs in each factor.

Fix $\ell \geq 1$ and variables $z_1, \dots, z_\ell$. For $I = (i_1, \dots, i_\ell) \in \mathbb{Z}_{\geq 0}^\ell$, put $S := \sum_c i_c$, $A_c := t^{i_c}$, $K_I := \prod_{c < c'} K_{i_c i_{c'}}(z_c, z_{c'})$, and

$$V_I^{(n)} := K_I \sum_{\sum n_c = n} s^{(\ell-1)\sum_c (n_c - i_c)} t^{-\sum_{c \neq c'} (n_c - i_c) i_{c'}} \prod_c N_{i_c}^{(n_c)} z_c^{-n_c},$$

with $V_I := 0$ if some $i_c < 0$.

Theorem 1 ($\star\ell$). For all $m, k \geq 0$, in $\Lambda_m \otimes \mathbb{Q}(s, t)(z)$:

$$\Gamma_k := \sum_a \prod_c z_c^{a_c} t^{-\binom{k}{2}} e_k(Y) \bullet (e_{a_1} \cdots e_{a_\ell}) = T_k := \sum_{b, I} (s^\ell t^{-S})^b V_I^{(k-b)} \cdot e_b \prod_c E(t^{i_c} z_c).$$

Writing $n = \sum n_c$, this is PROVE.md’s formula verbatim. The special cases:

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- $\ell = 1$ gives 207b's $e_k \star e_r$;
- $\ell = 2$ gives (TC) (209).

Corollary 2. Γ_k is a polynomial in z . Hence the coefficient of z^a in T_k is $e_k \star (e_{a_1} \cdots e_{a_\ell})$ in Hikita's reading, exactly as in 209. Because the e_a generate, this gives $e_k \star F$ for every $F \in \Lambda$.

2. THE RECURSION (Rℓ)

This is 209 §1 verbatim with $\varphi(u) := u \prod_c (1 + suz_c)/(1 + uz_c)$. 207b's (A_k) , (K_k) and (R) need only a symmetric F , and $e_{a_1} \cdots e_{a_\ell}$ is symmetric. The generating function of $X_A \prod_c e_{a_c}(X_{A^c}, sX_A)$ is $X_A \prod_c \prod_{A^c} (1 + Xz_c) \prod_A (1 + sXz_c)$. This gives

$$(R\ell) \quad [k] \Gamma_k(X) = \sum_i X_i \prod_c (1 + sX_i z_c) \prod_{j \neq i} a_{ij} \Gamma_{k-1}(\hat{X}_i) \quad (m \geq 1),$$

with $\Gamma_k = 0$ for $m = 0$ and $k \geq 1$, and $\Gamma_0 = \prod_c E(z_c)$. This recursion is the one implemented in `scripts/day210/gf_recursion_ell.py`.

3. STEP MAP

Fix one term $\mathcal{W} e_b(\hat{X}_i) \prod_c E(\hat{X}_i; \gamma_c)$ of $T_{k-1}(\hat{X}_i)$, with $\gamma_c = t^{ic} z_c$. As in 209 §2, it contributes

$$\mathcal{W} \prod_c E(\gamma_c) \cdot [x^b] \sum_i E(x) H(X_i; x) \prod_{j \neq i} a_{ij}, \quad H(y; x) := \frac{y \prod_c (1 + syz_c)}{\prod_c (1 + \gamma_c y) \cdot (1 + xy)}.$$

The poles of H/y are simple, at $-1/a$ for $a \in \{x, \gamma_1, \dots, \gamma_\ell\}$. At infinity $H/y \sim c/(xy)$, where

$$c := s^\ell \prod z_c / \prod \gamma_c = s^\ell t^{-S}.$$

At $y = -1/a$ the residue of $H(y)Q(1/y)/y$ is $\rho_a \cdot E(ta)/E(a)$, with

$$\rho_a := \frac{\prod_c (a - sz_c)}{a \prod_{a' \neq a} (a - a')}.$$

To see this, note that $\prod_c (1 - sz_c/a) = \prod (a - sz_c)/a^\ell$ and $\prod_{a' \neq a} (1 - a'/a) = \prod (a - a')/a^\ell$, and that $Q(-a) = E(ta)/E(a)$. Lemma 2' (209 §2) then gives

$$(1 - t) \sum_i H(X_i; x) \prod a_{ij} = \frac{c}{x} - \sum_{a \in \{x, \gamma\}} \rho_a \frac{E(ta)}{E(a)}.$$

Now let $F(y) := \prod_c (y - sz_c)/(y \prod_c (y - \gamma_c))$. Then $\rho_x = F(x)$, and:

- $\text{Res}_0 F = c$;
- $\text{Res}_{\gamma_c} F = \kappa_c := \frac{\prod_{c'} (\gamma_c - sz_{c'})}{\gamma_c \prod_{c' \neq c} (\gamma_c - \gamma_{c'})}$;
- $\rho_{\gamma_c} = -\kappa_c/(x - \gamma_c)$.

Every term is $O(1/x)$ at $x = \infty$. Extracting the coefficient of e_n (argued exactly as in 209 §2) gives the **step map**:

- $n = b + 1$: $c(1 - t^{b+1}) \prod E(\gamma_c)$.
- $n \leq b$: $-\sum_c \kappa_c \gamma_c^{n-b-1} (E(t\gamma_c) - t^n E(\gamma_c)) \prod_{c' \neq c} E(\gamma_{c'})$.
- $n > b + 1$: 0.

Machine checks:

- `step_map_ell.py` compares this step map with the Day 210 recursion output ($\ell = 3$, $k \leq 2$, symbolic);
- `step_map_point.py` iterates the step map from Γ_0 and compares with T_k at exact points ($\ell \leq 5$; see §7).

4. STEP LEMMA (L ℓ) AND REDUCTION TO ($\star\ell$ -GF)

Lemma 3 (L ℓ). $\sum_i X_i \prod_c (1 + sX_i z_c) \prod_{j \neq i} a_{ij} T_{k-1}(\widehat{X}_i) = [k] T_k.$

Write $W'_{bI} = (s^\ell t^{-S})^b V_I^{(k-1-b)}$. By §3, the coefficient of $e_{b'} \prod E(t^{i_c} z_c)$ in $(1-t) \cdot \text{LHS}$ is

$$W'_{b'-1, I} c(1-t^{b'}) + \sum_{b \geq b'} \left[t^{b'} W'_{bI} \sum_c \kappa_c \gamma_c^{b'-b-1} - \sum_c W'_{b, I-e_c} \kappa'_c (t^{i_c-1} z_c)^{b'-b-1} \right],$$

where κ'_c is κ_c evaluated at the γ 's of $I - e_c$. Put $n = k - b'$ and $p = b - b'$, and factor out $P := (s^\ell t^{-S})^{b'}$:

- $W'_{b'-1, I} c = P \cdot V_I^{(n)}$;
- $W'_{bI} = P \cdot c^p V_I^{(n-1-p)}$;
- $W'_{b, I-e_c} = P \cdot t^{b'} (ct)^p V_{I-e_c}^{(n-1-p)}$, since the c -value of $I - e_c$ is ct .

So (L ℓ) follows, exactly as in 209 §3, from

$$(\star\ell) \quad \sum_{p \geq 0} c^p \left[V_I^{(n-1-p)} \sum_c \kappa_c \gamma_c^{-p-1} - t^p \sum_c V_{I-e_c}^{(n-1-p)} \kappa'_c (t^{i_c-1} z_c)^{-p-1} \right] = (1-t^n) V_I^{(n)}.$$

Given ($\star\ell$), the coefficient is $P(1-t^{b'+n})V_I^{(n)} = (1-t^k) \cdot (\text{the } T_k \text{ weight})$. ($\star\ell$) and $T_0 = \Gamma_0$ then prove the Theorem by induction on k for all m at once, as 209 §3 does, and (R ℓ) supplies the $m \geq 1$ step.

Put $\mathcal{V}_I(x) := \sum_n V_I^{(n)} x^n$. The p -sums are convolutions, and $\sum_p c^p \gamma^{-p-1} x^{p+1} = x/(\gamma - cx)$. So ($\star\ell$) for all n is equivalent to

$$(\star\ell\text{-GF}) \quad \mathcal{V}_I(x)U(x) - \mathcal{V}_I(tx) = - \sum_c \kappa'_c \frac{x}{t^{i_c-1} z_c - ctx} \mathcal{V}_{I-e_c}(x), \quad U := 1 - \sum_c \frac{\kappa_c x}{\gamma_c - cx}.$$

5. PROOF OF ($\star\ell$ -GF)

Closed form of $\mathcal{V}_I$ (**P5**). Let $X_c := s^{\ell-1} t^{-S} x/z_c = cx/(sz_c)$. The t -exponent attached to column c is $-(n_c - i_c)(S - i_c) - n_c i_c = -n_c S + i_c(S - i_c)$. The n_c -sums therefore factor:

$$\mathcal{V}_I(x) = K_I s^{-(\ell-1)S} t^{S^2 - \sum i_c^2} \prod_c C_{i_c}(X_c).$$

For $I - e_c$, every $X_{c'}$ becomes $tX_{c'}$.

Step A: U factors (**P1**)–(**P3**).

- F has numerator degree ℓ and denominator degree $\ell + 1$, so $F = c/y + \sum_c \kappa_c/(y - \gamma_c)$. (**P1**)
- Hence $U = xF(cx)$. (**P2**)
- We have $cx - sz_c = -sz_c(1 - X_c)$ and $cx - \gamma_c = -\gamma_c(1 - sX_c/A_c)$. With $c \prod \gamma_c = s^\ell \prod z_c$ this gives

$$(\text{P3}) \quad U = \prod_c \frac{1 - X_c}{1 - sX_c/A_c}.$$

Step B: normalise. Divide ($\star\ell$ -GF) by $\mathcal{N} := K_I s^{-(\ell-1)S} t^{S^2 - \sum i_c^2} \prod_c D_{i_c} X_c^{i_c} G(t^2 X_c)$. This is legitimate because $D_j \neq 0$ (207b §4). 207b gives $C_i(X) = D_i X^i G(tX)(1 - sX/A)/(1 - X)$ and $G(tX)/G(t^2 X) = (1 - stX)/(1 - tX)$. With $u_c := stX_c$:

- **Left side.** Column c of $\mathcal{V}_I \cdot U$ gives $\frac{C_i(X)}{D_i X^i G(t^2 X)} \cdot \frac{1-X}{1-sX/A} = \frac{1-stX}{1-tX} = \frac{1-u}{1-tX}$. Column c of $\mathcal{V}_I(tx)$ gives $\frac{C_i(tX)}{D_i X^i G(t^2 X)} = \frac{A(1-stX/A)}{1-tX} = \frac{A-u}{1-tX}$. So

$$L_0 := \frac{\prod_c (1-u_c) - \prod_c (A_c - u_c)}{\prod_c (1-tX_c)}.$$

- **Shifted term c ($i_c \geq 1$).** The factors are:

- (P4) $\frac{x}{t^{i_c-1} z_c - ct x} = \frac{stX_c}{c(A_c - st^2 X_c)}$;
- the prefactor ratio of $\mathcal{V}_{I-e_c}$ to $\mathcal{N}$ is $s^{\ell-1} t^{(S-1)^2 - \sum i'^2 - S^2 + \sum i^2} = s^{\ell-1} t^{-2(S-i_c)}$;
- for columns $c' \neq c$: $C_{i_{c'}}(tX_{c'})/(\dots) = (A_{c'} - u_{c'})/(1-tX_{c'})$;
- for column c : $C_{i_c-1}(tX_c)/(\dots) = \frac{1-A_c}{ts-A_c} \cdot \frac{A_c}{t} X_c^{-1} \frac{1-st^2 X_c/A_c}{1-tX_c}$, from $D_{i-1}/D_i = (1-A)/(ts-A)$;
- (P6) κ' splits pairwise: $\kappa'_c = \frac{\gamma'_c - sz_c}{\gamma'_c} \cdot \prod_{c' \neq c} \frac{\gamma'_c - sz_{c'}}{\gamma'_c - \gamma_{c'}}$. K_{I-e_c}/K_I is a product of the $\ell-1$ pair ratios involving c . Using $K_{ij}(z, w) = K_{ji}(w, z)$ (§1), we may put c in the first slot of each pair. Then the pair factor of κ' times the pair ratio of K is exactly the 209 (P6) computation, with $(z, w, A, B) \rightarrow (z_c, z_{c'}, A_c, A_{c'})$. The remaining diagonal factor is $(\gamma'_c - sz_c)/\gamma'_c = (A_c - st)/A_c$. With $\rho_{cc'} := z_c/z_{c'} = X_{c'}/X_c$:

$$\Phi_c := \kappa'_c K_{I-e_c}/K_I = \frac{A_c - st}{A_c} \cdot \prod_{c' \neq c} \frac{A_{c'}(A_c \rho_{cc'} - 1)}{A_c \rho_{cc'} - A_{c'}}.$$

Multiply these together:

- $\frac{stX_c}{c(A_c - st^2 X_c)} \cdot \frac{A_c}{t} X_c^{-1} (1 - st^2 X_c/A_c) = s/c$;
- $s \cdot s^{\ell-1}/c \cdot t^{-2(S-i_c)} = t^{2i_c-S} = A_c/\prod_{c' \neq c} A_{c'}$;
- $(A_c - st)/(ts - A_c) = -1$.

The shifted term, moved to the left, is therefore $S_c/\prod(1-tX)$, where

$$S_c = (1 - A_c) \Psi_c \prod_{c' \neq c} (A_{c'} - u_{c'}), \quad \Psi_c := - \prod_{c' \neq c} \frac{A_c X_{c'} - X_c}{A_c X_{c'} - A_{c'} X_c}.$$

For $i_c = 0$ the term is absent, and the formula gives 0 anyway because of the factor $1 - A_c$. For $\ell = 2$ this reproduces 209's S_1 and S_2 . ($\star\ell$ -GF) is now equivalent to $\prod(1-u_c) - \prod(A_c - u_c) + \sum_c S_c = 0$.

Step C: the closing identity. Put $\lambda_c := A_c/u_c$ and $\mu_c := 1/u_c$. Then:

- $A_{c'} - u_{c'} = u_{c'}(\lambda_{c'} - 1)$;
- $(A_c X_{c'} - X_c)/(A_c X_{c'} - A_{c'} X_c) = (\lambda_c - \mu_{c'})/(\lambda_c - \lambda_{c'})$, since $X \propto u$;
- $(1 - A_c)/u_c = \mu_c - \lambda_c$.

Divide by $\prod u_c$. The identity becomes

$$(Z) \quad \prod_c (\mu_c - 1) - \prod_c (\lambda_c - 1) = \sum_c (\mu_c - \lambda_c) \prod_{c' \neq c} \frac{(\lambda_{c'} - 1)(\lambda_c - \mu_{c'})}{\lambda_c - \lambda_{c'}},$$

for free λ, μ with the λ_c distinct and $\neq 1$.

Proof of (Z). Let $g(w) := \prod_c (w - \mu_c)/((w-1) \prod_c (w - \lambda_c))$. Its residues are:

- at $w = \lambda_c$: $\frac{\lambda_c - \mu_c}{\lambda_c - 1} \cdot \prod_{c' \neq c} \frac{\lambda_c - \mu_{c'}}{\lambda_c - \lambda_{c'}}$;
- at $w = 1$: $\prod(1 - \mu_c)/\prod(1 - \lambda_c) = \prod(\mu_c - 1)/\prod(\lambda_c - 1)$;

- at ∞ : -1 , since $g \sim 1/w$.

The residues sum to 0. Multiply by $\prod_c(\lambda_c - 1)$. The residue at λ_c becomes minus the c -th summand on the right of (Z), and (Z) follows. $\square$

Remark. (Z) is almost certainly a classical partial-fractions/residue (Lagrange-interpolation) identity, and no novelty is claimed for it. The claim of this note is the rule $(\star\ell)$ (Theorem 1), not (Z). Nor is any novelty inferred from the pairwise shape of the cross kernel K_I .

Specialisation is legitimate. As rational functions in x , the $\lambda_c = A_c/(stX_c)$ are distinct: $\lambda_c = \lambda_{c'}$ would force $t^{ic}z_c = t^{ic'}z_{c'}$. They are also $\neq 1$. Nothing else was divided by: $1 - sX_c/A_c$, $ts - A_c$ and D_j are nonzero. So $(\star\ell\text{-GF})$ holds, which gives $(\star\ell)$ for every n , which proves the Theorem. $\square$

6. WHAT MADE IT SHORT

- At $\ell = 2$, (P7) “ $u/X = v/W$ ” looked like luck. It is (Z) at $\ell = 2$: a Lagrange-interpolation identity at the nodes $\lambda_c = A_c/u_c$, with the “zeros” $\mu_c = 1/u_c$ and the extra pole at $w = 1$.
- Two residue functions carry the whole proof:
 - $F(y) = \prod(y - sz_c)/(y \prod(y - \gamma_c))$ gives the step map and U ;
 - $g(w) = \prod(w - \mu_c)/((w - 1) \prod(w - \lambda_c))$ closes it.
- The pairwise cross kernel K is forced by the pairwise splitting of κ' .

7. VERIFICATION (`SCRIPTS/DAY212/`)

All logs ALL OK.

check	scope	log
step map (§3) = Day 210	$\ell = 3, k \leq 2$, symbolic	<code>step_map_ell3.log</code>
<code>gf_recursion_ell</code>		
iterated step map = T_k , symbolic	$\ell = 3, k \leq 2$ (larger runs too slow in sympy; see next row)	<code>step_map_ell3.log</code>
iterated step map = T_k at 2 exact rational points	$\ell \leq 3, k \leq 6$; $\ell = 4, k \leq 5$; $\ell = 5, k \leq 4$ (54/54)	<code>step_map_point.log</code>
(Z)	$\ell \leq 5$, free λ, μ , symbolic	<code>check_steps.log</code>
normalised $(\star\ell\text{-GF})$, free A_c	$\ell \leq 4$, symbolic	<code>check_steps.log</code>
(P1)–(P4)	$\ell \leq 4$, free A_c , symbolic	<code>check_steps.log</code>
(P5), (P6), $(\star\ell)$ coefficient form	$\ell \leq 4$, all I with $\Sigma I \leq 4$ ($\ell = 4$: ≤ 3), $n \leq 6$, 2 exact points	<code>check_concrete.log</code>
Step B composite: shifted term/ $\mathcal{N} = S_c$, L_0 formula, total = 0	$\ell = 2, 3, 4$, all $I \in [0, 2]^\ell$, symbolic in x	<code>check_stepB.log</code>
end-to-end vs direct AHA (Day 210)	$\ell = 3, m \leq 6, k \leq 5$; $\ell = 4, m = 4, k \leq 3$	<code>scripts/day210/aha_check_ell3*.log</code>

8. SCOPE, GAPS, CREDITS

Gaps. None in the argument. The load-bearing inputs are:

- 207b (A_k), (K_k), (R) and the §4 facts about C_i and D_j (PROVED; Clio peer-verified the $\ell = 1$ result);
- Lemma 2' (209);
- the residue theorem.

Hikita reading. This relies on Hikita 2503.23597 Def. 3.4 / Lemma 3.3, as in 207b and 209.

Not done.

- A positivity or DS extraction at general length.
- An explicit bounded formula for $e_k \star e_\lambda$.

Next.

- Extract $e_k \star e_\lambda$ coefficients.
- Check whether (Z) is the “column-exchange” shadow of a known Lagrange identity: it smells like Milne’s $U(n)$ or the Gustafson residue lemma. This needs a novelty audit.